



Department of Computer Science and Engineering
Midterm Examination Fall 2024
CSE 321: Operating Systems

Duration: 1 Hour 20 Minutes

Total Marks: 30

Answer the following questions.
 Figures in the right margin indicate marks.

1. CO1	a) Howard, a brilliant engineer, got into NASA. He was tasked with developing an operating system for the next rover that NASA is going to send to Mars. The program director gave Howard the following requirements: • The OS components communicate with each other using messages. • The programmers can easily add new features to the OS. • The kernel has limited operations. Figure out the OS structure to be used according to the scenario. State proper reasons.	[3]																		
	b) As a part of designing your operating system, you finally face the daunting task of designing a scheduler. Your friend suggests a round robin algorithm (time quantum = 5s) to implement your scheduler. However, you decide to go with a preemptive version of priority scheduler where the priority of a process is inversely proportional to its burst period. You are given the following arrival burst periods of five processes: <table border="1" style="margin-left: auto; margin-right: auto;"> <thead> <tr> <th>Process</th><th>Arrival Time (s)</th><th>Burst Time (s)</th></tr> </thead> <tbody> <tr> <td>P1</td><td>0</td><td>10</td></tr> <tr> <td>P2</td><td>0</td><td>9</td></tr> <tr> <td>P3</td><td>2</td><td>4</td></tr> <tr> <td>P4</td><td>5</td><td>6</td></tr> <tr> <td>P5</td><td>7</td><td>4</td></tr> </tbody> </table>	Process	Arrival Time (s)	Burst Time (s)	P1	0	10	P2	0	9	P3	2	4	P4	5	6	P5	7	4	
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	I. Draw Gantt Charts for above mentioned algorithms. II. Calculate average waiting times and turnaround times for both algorithms.	[6+6] [3+3]																		

2. CO2	a) Find outputs of the given code: <pre> int main(){ pid_t a,b; double x=15.0; double y=11.5; double z=5.25; a=fork(); if(a<0){ printf("fork failed\n"); } else if(a==0){ x=(x+y)-z; y=(x-y)+z; z=x+y+z; b=fork(); if(b<0){ printf("fork failed\n"); } else if(b>0){ printf("Child Starts\n"); wait(); x=x+y; y=y+z; z=x+z; } else{ printf("Grand Child Starts\n"); x=x+y; y=y+z; z=x+z; } } else{ printf("Parent Starts\n"); wait(); x=x*y; y=x/y; z=(z*y)/x; } printf("x = %lf\n",x); printf("y = %lf\n",y); printf("z = %lf\n",z); return 0; } </pre>	[6]
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	b) Alex wants to develop such a platform where users from around the world can share information among themselves. His motive is to connect users under a server and use a dedicated storage to exchange information. In order to share information users require to send and receive data in the storage among each other. For his platform Alex is more concerned about the size of the storage over data transfer rate. Identify with proper logic which communication method is suitable for Alex's platform?	[3]
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